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Course structure

1. Transport greenhouse gas emissions
2. Origin–destination vs destination mobility
3. The rationale for flying
4. Airport access and egress
5. Environmental policy instruments
6. Sustainable aviation
7. Green airports
8. High-speed rail
9. Mobility as a Service

How the practical sessions work

- Start with a transport problem and define its assumptions.
- Change parameters in the Streamlit simulator.
- Interpret magnitudes, compare scenarios and discuss limitations.
- Distinguish illustrative calculations from empirical estimates.

1. Transport greenhouse gas emissions

Concepts and interactive practice

What is measured?

- CO_2 and other greenhouse gases can be expressed as CO_2 equivalents.
- Organisational boundaries and operational scopes determine what is included.
- Compare total emissions and emissions per passenger-kilometre.

GHG accounting boundaries

- Scope 1: direct emissions from owned or controlled sources.
- Scope 2: emissions associated with purchased energy.
- Scope 3: other indirect value-chain emissions; avoid double-counting within an inventory.

Activity-based accounting

- Emissions = activity data $\times$ relevant emission factor.
- Document factor units, year, system boundary and uncertainty.
- Different modes require consistent occupancy and lifecycle assumptions.

Classroom exercise

- Set activity levels and emission factors in the app.
- Calculate emissions by scope and compare a baseline with a reduction scenario.

2. Tourism transport: O/D and destination mobility

Concepts and interactive practice

Two mobility components

- Origin–destination mobility links home and tourism destination.
- Intra-destination mobility describes travel after arrival.
- Their policy levers and accounting boundaries differ.

Tourism gravity model

- $T_{ij} = kP_i^{\alpha} A_j^{\gamma} / D_{ij}^{\beta}$
- Population, attractiveness and distance enter through chosen elasticities.
- Calibrating a single observation identifies k only when elasticities are fixed.

Interpreting gravity results

- A 1% increase in P_i changes predicted flows by approximately α%.
- Changing D_{ij} produces an inverse effect governed by β.
- Predictions are conditional on model assumptions, not causal estimates.

Classroom exercise

- Calibrate the model on an illustrative route.
- Predict passenger flows after changing distance or destination appeal.

3. The rationale for flying

Concepts and interactive practice

Why do people fly?

- Travel time, price, convenience and available alternatives matter.
- Habits and perceived environmental consequences may also matter.
- Attitudes and behaviour need not coincide.

Discrete choice representation

- $U_{im} = V_{im} + \varepsilon_{im}$
- $P_{im} = e^{V_{im}} / \sum_j e^{V_{ij}}$
- Coefficients in the simulator are pedagogical assumptions, not estimated preferences.

Classroom exercise

- Compare modal choice probabilities when airfares and rail times change.
- Identify assumptions needed to infer real demand from surveys.

4. Airport access and egress

Concepts and interactive practice

Access is part of the journey

- Access: travel to the departure airport.
- Egress: travel from the arrival airport.
- Reliability, luggage and transfers can affect perceived convenience.

Generalised travel cost

- $GC_m = C_m + VOT\ t_m + R_m$
- Monetary costs, time valuation and other penalties use consistent units.
- Travel time includes transfers, waiting and access legs if relevant.

Airport accessibility

- Catchment areas depend on travel impedance and competing airports.
- Compare scenarios using generalised costs rather than distance alone.

Classroom exercise

- Vary fares, travel time and the value of time for bus, taxi and rail.
- Find the lowest-cost mode under alternative passenger profiles.

5. Environmental policy measures

Concepts and interactive practice

Market-based instruments

- An emissions tax prices pollution per tonne.
- Emissions trading fixes the quantity of allowances (subject to scheme design).
- A carbon price changes financial incentives; effects depend on response.

Carbon price arithmetic

- Carbon cost $= E_{tCO_2} \times p_{\mathrm{EUR}/tCO_2}$
- Use an explicit baseline, coverage assumptions and pass-through rate.
- A tax calculation alone does not estimate behavioural change.

Classroom exercise

- Set emissions and carbon prices.
- Explore changes in total payment and assumed fare pass-through.

6. The agenda for sustainable aviation

Concepts and interactive practice

Routes to lower emissions

- Operational efficiency affects fuel use per flight.
- Sustainable aviation fuels may reduce lifecycle emissions depending on pathway.
- Demand growth can offset efficiency improvements.

Scenario accounting

- $E_1 = E_0(1+g)(1-r)$
- Here g is passenger activity growth and r is intensity reduction.
- This simple identity does not include non-CO_2 climate effects.

Classroom exercise

- Compare growth and efficiency assumptions through 2050.
- Test whether absolute emissions rise or fall.

7. Green airports

Concepts and interactive practice

Airport carbon management

- Inventory terminal electricity and direct airport operations.
- Distinguish airport-controlled sources from aircraft operators' sources.
- Measures include equipment electrification and electricity procurement.

Airport Carbon Accreditation

- Levels cover progressively broader measurement and management requirements.
- An accreditation level is not a measure of the total climate impact of flying.

Classroom exercise

- Construct an illustrative airport emissions inventory.
- Compare electricity, vehicles and equipment interventions.

8. High-speed rail and integration

Concepts and interactive practice

Competition and integration

- Compare door-to-door journeys, not in-vehicle time alone.
- Rail–air integration requires schedules, ticketing and transfer arrangements.
- High-speed rail may replace or complement air connections.

Modal competition model

- $V_m = -\beta_c C_m - \beta_t t_m$
- $P_m = e^{V_m} / \sum_j e^{V_j}$
- Numerical parameters are illustrative and are not estimated from data.

Classroom exercise

- Change train and air prices, times and first/last-mile costs.
- Compare predicted shares and discuss model limitations.

9. Mobility as a Service

Concepts and interactive practice

What MaaS integrates

- Information, journey planning, booking, payment and travel entitlements.
- Packages may combine collective and shared mobility options.
- Operators, public authorities and platform providers have different roles.

Bundles and user costs

- $GC =$ subscription $+$ out-of-bundle charges $+ VOT \times t$
- Compare bundles with individually purchased trips.
- Financial savings and reductions in vehicle travel are different outcomes.

Classroom exercise

- Build a monthly tourist mobility bundle.
- Test break-even usage and compare individual ticket costs.

Discussion

- Which parameter change was most consequential?
- Which assumptions could be tested empirically?
- What additional data would a policymaker need?